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Pharmacology: Antidepressants and Mood Stabilizers

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Session Objectives

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1. Describe the various hypotheses for depression.
2. Contrast the various groups of antidepressants. Differentiate between their MOA. Include new approvals.
3. Describe adverse effects and their management.
4. Describe bipolar disorder and the specific treatments.
5. Describe lithium and its adverse effects.

Source: Course Syllabus

Amine Hypothesis

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- Reserpine depletes DA, NE, and 5-HT. This drug is associated with depression.
- Most drugs to treat depression increase these neurotransmitters.
- Beta-adrenergic receptor blockers and $alpha_2$ agonists may alter mood.

Source: Course Syllabus

Other

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- Steroids or those with increased cortisol have mood disorders.
- Estrogen and testosterone deficiency have increased depression.
- Abnormal thyroid function observed in depressed patients.
- Atrophy in hippocampus and cortex are associated with depression.
- Unipolar vs bipolar depression

Source: Course Syllabus

Antidepressants

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- Monoamine oxidase inhibitors (MAOIs)
- Tricyclic antidepressants (TCA)
- Selective serotonin reuptake inhibitors (SSRIs): first choice.
- Atypical antidepressants. New agents.
- Esketamine: treatment-resistant depression
- Weeks to have efficacy (boxed suicide warnings)
- Switching antidepressants: allow “washout” period? If MAOI allow 4-5 weeks.
- Elderly concerns?
- Pregnancy: fluoxetine, sertraline, citalopram (safest).
- Psychotherapy, ECT (non-drug)

Source: Course Syllabus

Monoamine Oxidase Inhibitors (MAOI)

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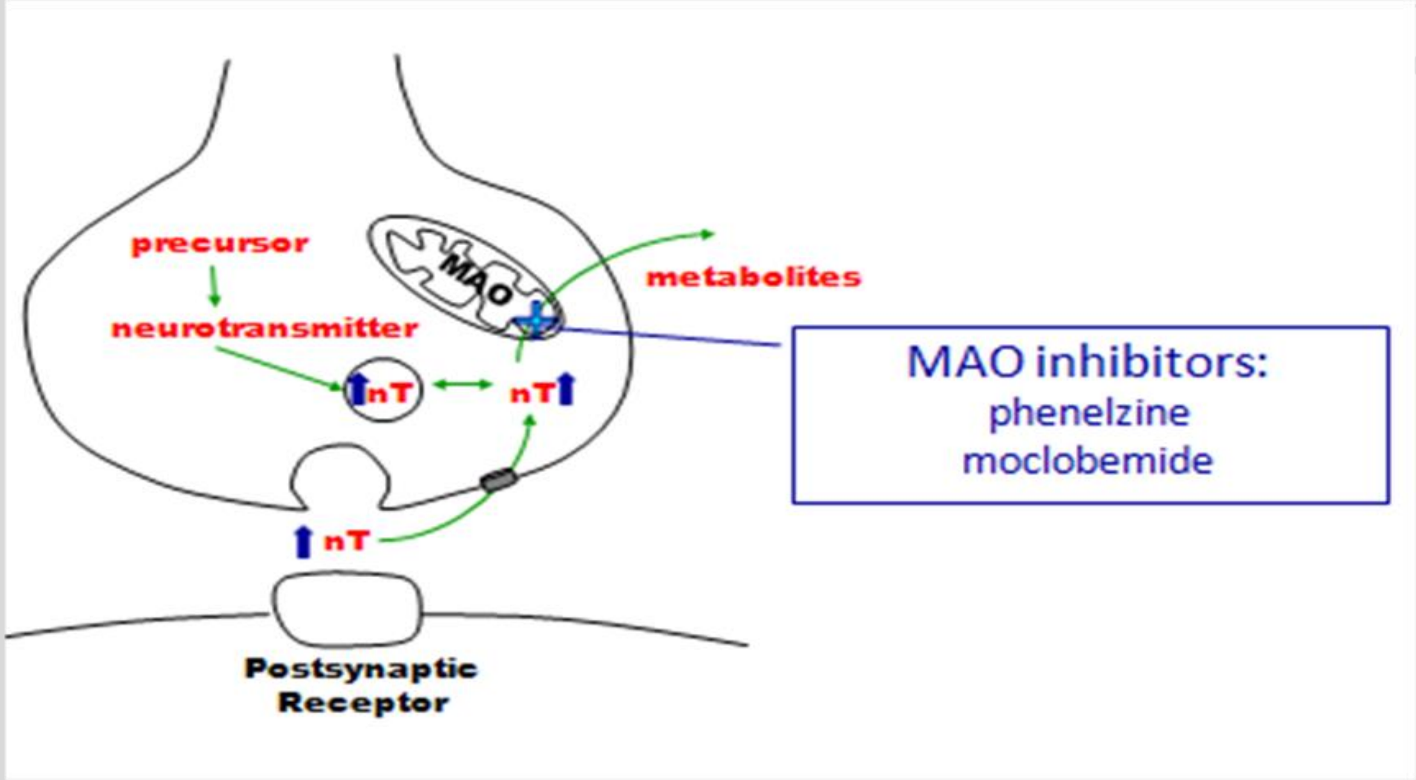
- MAO_A (NE, Epi, 5-HT, DA, tyramine)
- MAO_B (DA, Selegiline)
- Phenelzine, tranylcypromine, isocarboxazid
- Use: Refractory depression
- Drug and food interactions: Hypertensive crisis

Source: Course Syllabus

MAOI


precursor
neurotransmitter
MAO
metabolites
nT
nT
nT
MAO inhibitors:
phenelzine
moclobemide
Postsynaptic
Receptor

MAOI restrictions

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- Aged cheese and meats, red wine, chocolate, avocado, soy sauce, sauerkraut.
- Increased NE with alpha_1 agonists, sympathomimetics
- Serotonin syndrome (with meperidine, dextromethorphan, St John's wort, etc)
- Hypertensive crisis: Occipital headache, stiff neck, photophobia, palpitations, N/V

Source: Course Syllabus

Tricyclic Antidepressants (TCAs)

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- Imipramine: enuresis
- Desipramine: least anticholinergic
- Clomipramine: Obsessive-Compulsive Disorder (OCD)
- Amitriptyline: Neuropathic pain, migraine
- Nortriptyline
- Doxepin
- Amoxapine (some D_2 blockade). Adverse effects??

Source: Course Syllabus

TCAs

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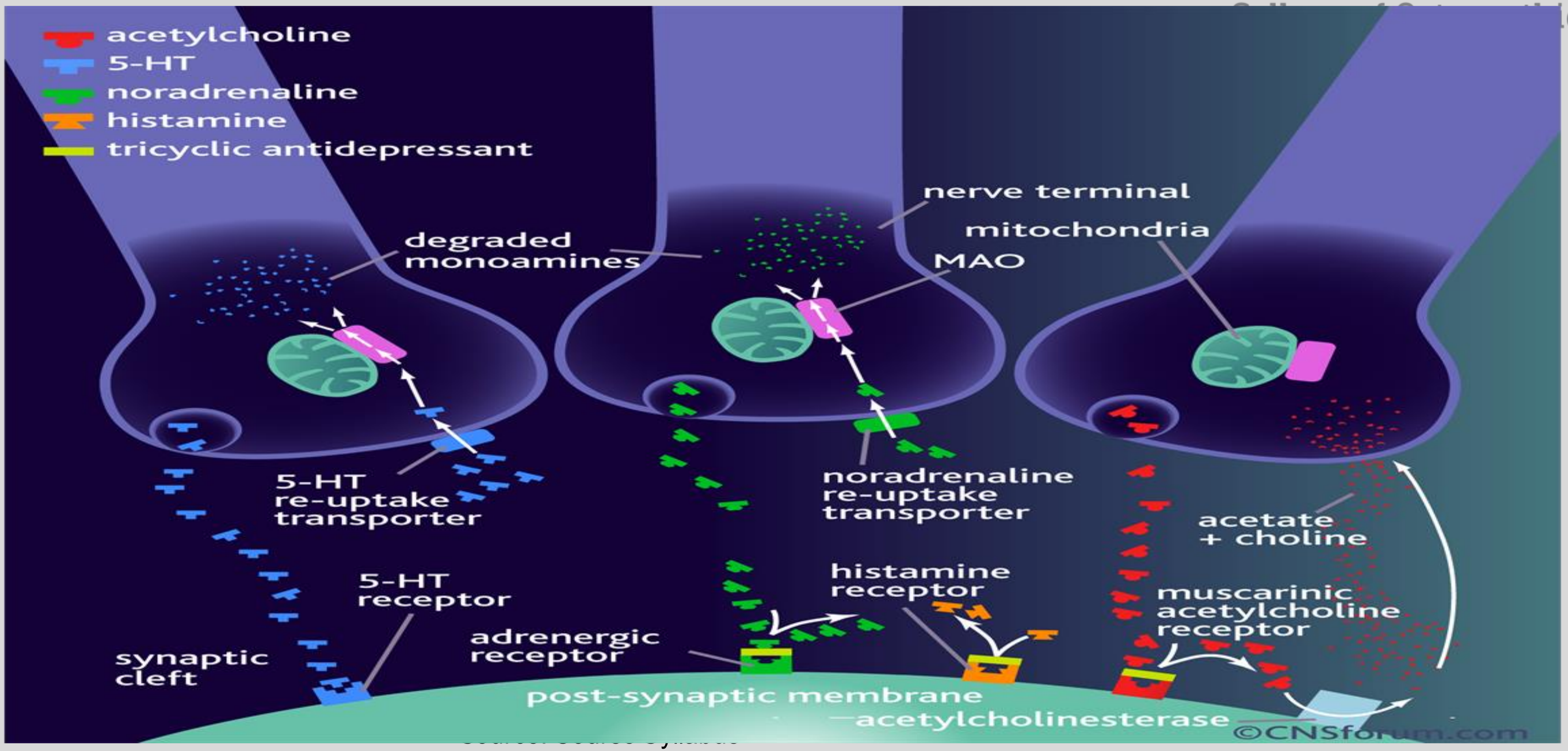
- Block NE and 5-HT reuptake (NET, SERT)
- Block alpha$_1$, M, and H$_1$ receptors: Name those adverse effects?
- CYP drug interactions. CYP2D6 substrate. Polymorphism is associated with slow metabolism of TCAs
- Serotonin Syndrome
- 3C's: coma, convulsions, cardiotoxicity ("quinidine-like"). Administer sodium bicarbonate.

Source: Course Syllabus

TCAs


acetylcholine
5-HT
noradrenaline
histamine
tricyclic antidepressant
nerve terminal
mitochondria
degraded monoamines
MAO
5-HT re-uptake transporter
noradrenaline re-uptake transporter
acetate + choline
5-HT receptor
histamine receptor
muscarinic acetylcholine receptor
adrenergic receptor
synaptic cleft
post-synaptic membrane
acetylcholinesterase
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Serotonin Syndrome

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- SSRIs, TCAs, meperidine, tramadol, dextromethorphan, St. John's Wort, linezolid, "triptans"
- <u>Symptoms:</u> rigidity, hyperthermia, hypertension, tachycardia, myoclonus, delirium, confusion, within 24 hrs, unlike NMS. Also, GI symptoms.
- <u>Treatment:</u> benzodiazepines, cyproheptadine, ventilation

Source: Course Syllabus

NMS vs Serotonin Syndrome

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	NMS	Serotonin Syndrome
Onset	Gradual (months-weeks)	Abrupt
Findings	“Lead-pipe” rigidity	Myoclonus and tremor
Reflexes	Decreased	Increased
Pupils	Normal	Mydriasis
Drugs	Antipsychotics, halogenated anesthetics	Linezolid, St. John’s Wort, “triptans”, SSRIs, MAOIs, tricyclic antidepressants

Serotonin Receptors (Summary)

https://www.psychiatrictimes.com/view/serotonin-a-biography

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Subtype	Receptor Mechanism	Drug	Clinical effects
$5HT_{1A}$	Gi	Buspirone (agonist)	Antianxiety Antidepressant Improved cognition
$5HT_{1B/1D}$	Gi	Sumatriptan (agonist)	Antimigraine
$5HT_{2A}$	Gq	LSD (agonist) Atypical Antipsychotics (antagonists)	Hallucinations (agonist) Antidepressant Decreases EPS
$5HT_{2c}$	Gq	Locaserin (agonist) Atypical Antipsychotics (antagonists)	Weight loss (agonist) Antidepressant,antianxiety, weight gain
$5HT_3$	Na^+/K^+ ion channel	Ondansetron (antagonist)	Antiemetic

Selective Serotonin Reuptake Inhibitors: (SSRIs)

- Fluoxetine (longest t ½)
- Fluvoxamine
- Paroxetine (not in pregnancy)
- Sertraline
- Citalopram (increase QT interval)
- Escitalopram

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Source: Course Syllabus

SSRIs

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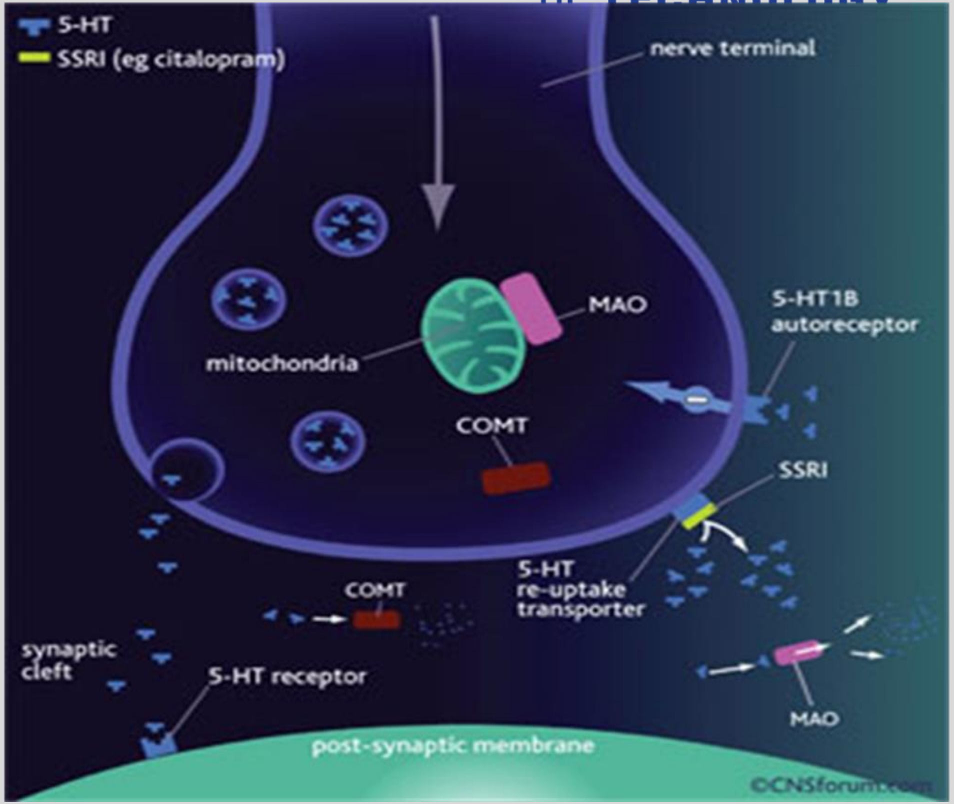
- Use: depression, OCD, panic disorder, bulimia, anorexia, PTSD, premenstrual dysphoric disorder
- 5-HT specific
- Fewer ANS and sedative effects than tricyclic antidepressants
- Safer in overdose (less cardio and seizures)



Source: Course Syllabus

SSRI Adverse effects

- Insomnia, headache
- Nausea, vomiting (activate $5\text{-}HT_3$)
- Discontinuation syndrome (least with fluoxetine)
- Bleeding abnormalities (platelets)
- Impotence (phosphodiesterase 5 inhibitor?)
- Serotonin syndrome
- Weight gain (block $5\text{-}HT_{2c}$)
- Drug interactions (most with fluoxetine CYP450 2D6; least with sertraline, citalopram, and escitalopram)

Source: Course Syllabus

Serotonin and norepinephrine reuptake inhibitors (SNRI)

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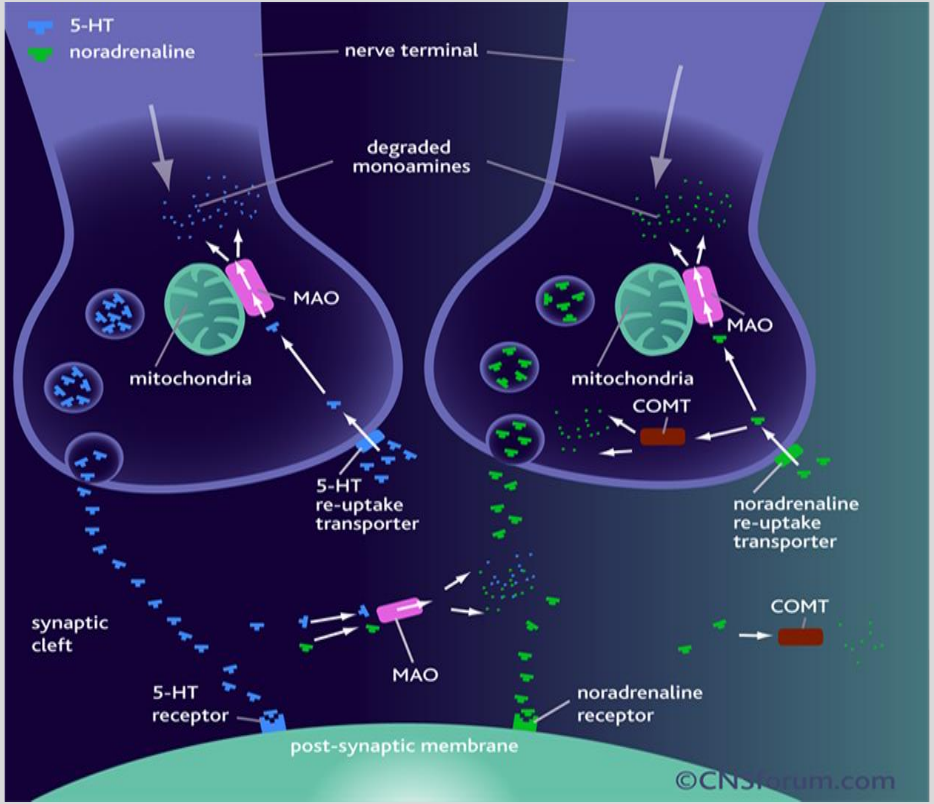
- No effect on H_1, alpha$_1$, M receptors, unlike TCA
- Use: Depression, fibromyalgia, neuropathic pain
- Duloxetine, venlafaxine, desvenlafaxine, milnacipran



Source: Course Syllabus

Bupropion

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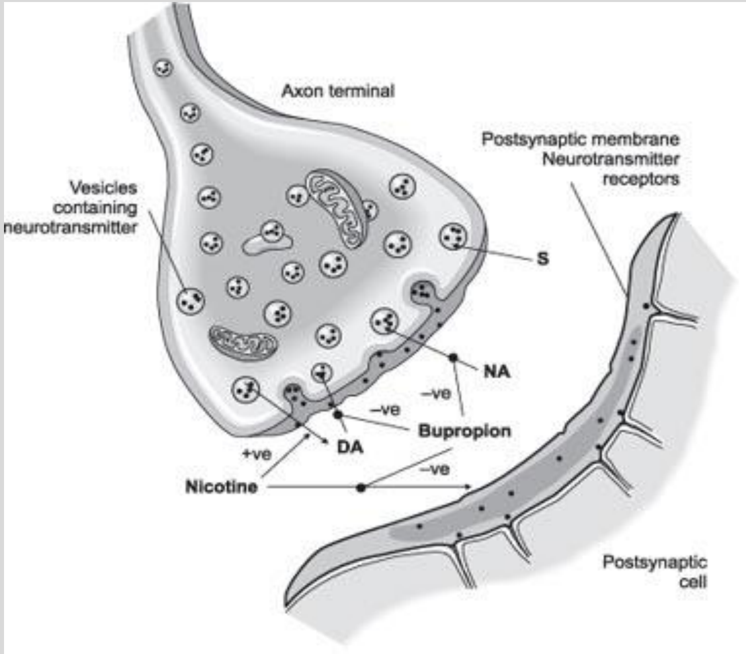
- Inhibit DAT, NET, inhibits dopamine and norepinephrine reuptake.
- Little M, H_1 effects
- No sexual effects
- CYP450 2D6 inhibitor
- Dextromethorphan/bupropion (Auvelity®, 2022)
- Naltrexone/bupropion (weight loss)
- Reduce seizure threshold (not for bulimia)
- Noncompetitive antagonist at nicotinic receptor (smoking cessation)
- Varenicline: partial nicotinic receptor agonist (adverse effects: erratic behavior, suicide)



Source: Course Syllabus

Mirtazapine

- Tetracyclic antidepressant (TeCA)
- Blocks alpha$_2$ to increase neuronal NE and 5-HT release
- Opposite clonidine
- Weight gain ($H_{1,}$ 5-HT$_{2c}$ blockade) Sedation (H_1 blockade)
- Blockade 5-HT$_{2A}$ (no sexual dysfunction)
- Blockade 5-HT$_3$ (no nausea)
- Stimulate 5-HT$_{1A}$ (anxiolytic)

Source: Course Syllabus

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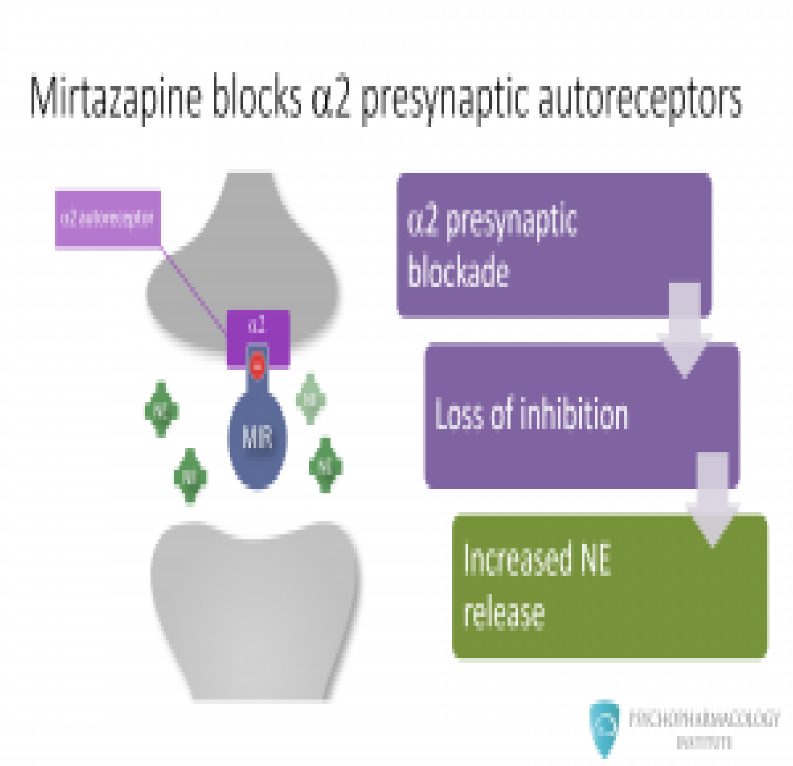

Trazodone

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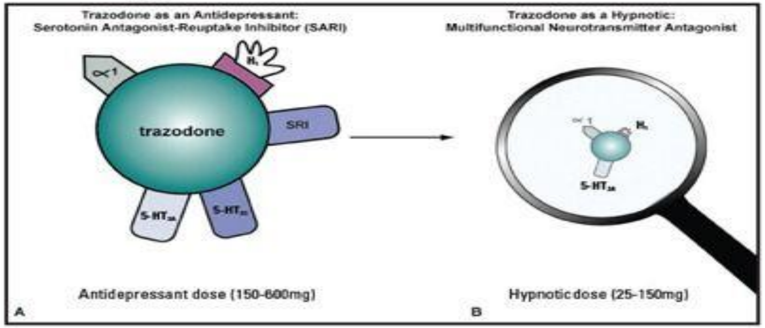
- Serotonin reuptake inhibitor and blocker (SARI) (5-HT_{2A})
- H_1 (sedation) and alpha$_1$ (priapism) blockade.
- Levels increased with CYP 3A4 inhibition
- Nefazodone: (hepatotoxic)
- Vilazodone (partial 5-HT_{1A} agonist).
- Vortioxetine: like vilazodone; also inhibits NET, SERT.

FIGURE 3.
Trazodone as an antidepressant and as a hypnotic



A) High doses that recruit saturation of SERT at 150–600 mg of trazodone are required for antidepressant actions. At this high antidepressant dose, trazodone is a multifunctional serotonergic agent with antagonist actions at 5-HT_{2A} and 5-HT_{2C} receptors as well. Thus, its antidepressant actions are attributed to these serotonergic properties and trazodone is a multifunctional SARI. It is also an α1 and H_1 antagonist at these doses.

B) Lower doses of trazodone in the 25–150 mg range lose saturation of SERT, and thus lose antidepressant actions while retaining multifunction antagonist actions at 5-HT_{2A}, H_1 and α1 receptors, and hypnotic efficacy.

α1=α1 adrenergic receptor; H_1=histamine 1 receptor; SRI=serotonin reuptake inhibitor; 5-HT_{2C}=serotonin 2C receptor; 5-HT_{2A}=serotonin 2A receptor; SERT=serotonin transporter; SARI=serotonin antagonist-reuptake inhibitor.

Stahl SM. *CNS Spectr*. Vol 14, No 10. 2009.

Source: Course Syllabus

Gepirone

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- **Selective 5-HT1$_A$ receptor agonist-- less binding to D$_2$ compared to buspirone.**
- **Less weight gain and sexual dysfunction.**
- **Use: Major depressive disorder, dosed once daily.**
- **<u>Adverse effects</u>: dizziness, headache, Increased QT prolongation**

Source: Course Syllabus

Brexanolone

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- Derivative of progesterone
- Modulate $GABA_A$ receptors (pre and post synaptic). Enhance hyperpolarization
- Rapid action, lasts 30 days
- Administered IV—Postpartum depression. Risk Evaluation and Mitigation strategy (REMS)
- Adverse effects: Sedation, dizziness
- Zuranolone: oral, No REMS needed, but no driving for 12 hours.

Bipolar disease

- **Bipolar I (at least one manic episode lasting at least 7 days, then depression for at least 2 weeks)**
- **Bipolar II (mainly depression; with episodes of hypomania lasting several days and more mild symptoms).**
- **Rapid-cycling: four or more episodes in a year.**
- **Treatments: lithium, valproate, carbamazepine, lamotrigine, or atypical antipsychotics**

Source: Course Syllabus

Lithium

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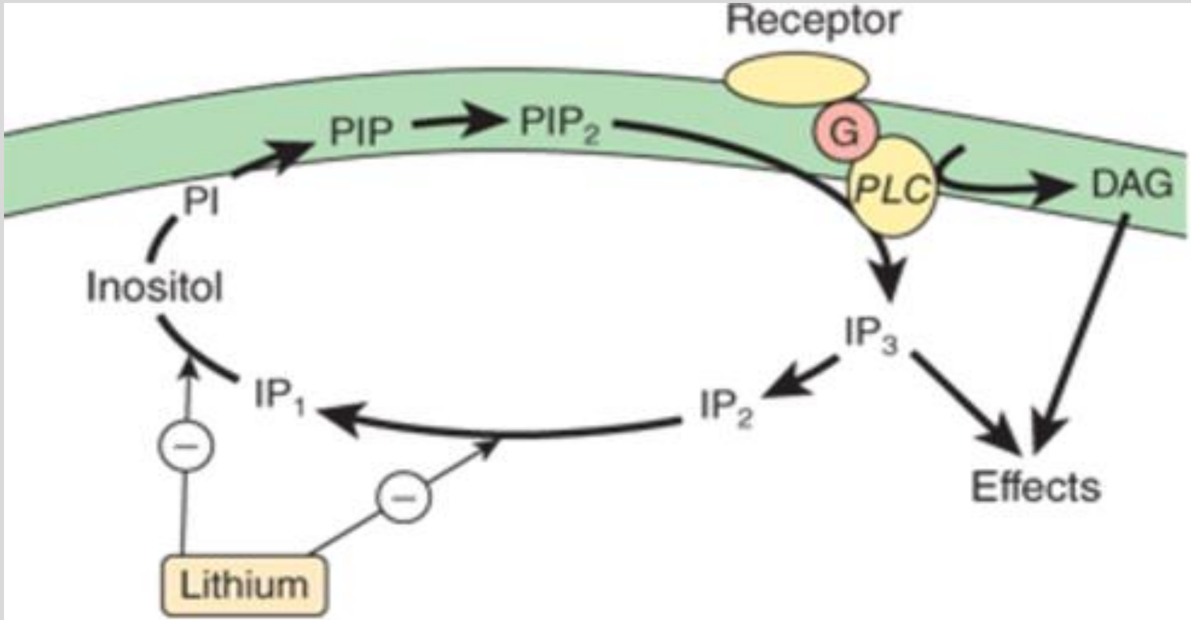
- Blocks inosine 5'monophosphatase (Gq), PKC
- Na+ loss promotes Li+ reabsorption (AE with diuretics)
- NSAIDs, ACEI facilitate Li+ reabsorption in proximal convoluted tubule (PCT).
- Narrow TI (1.5 mEq vs. 2mEq)
- Adverse effects: Tremors (use beta blocker)
- Leukocytosis (not related to infection)
- Polyuria, polydipsia (Loss of ADH response). Administer amiloride
- Hypothyroidism
- Ebstein anomaly (consider lamotrigine in pregnancy)

Source: Course Syllabus

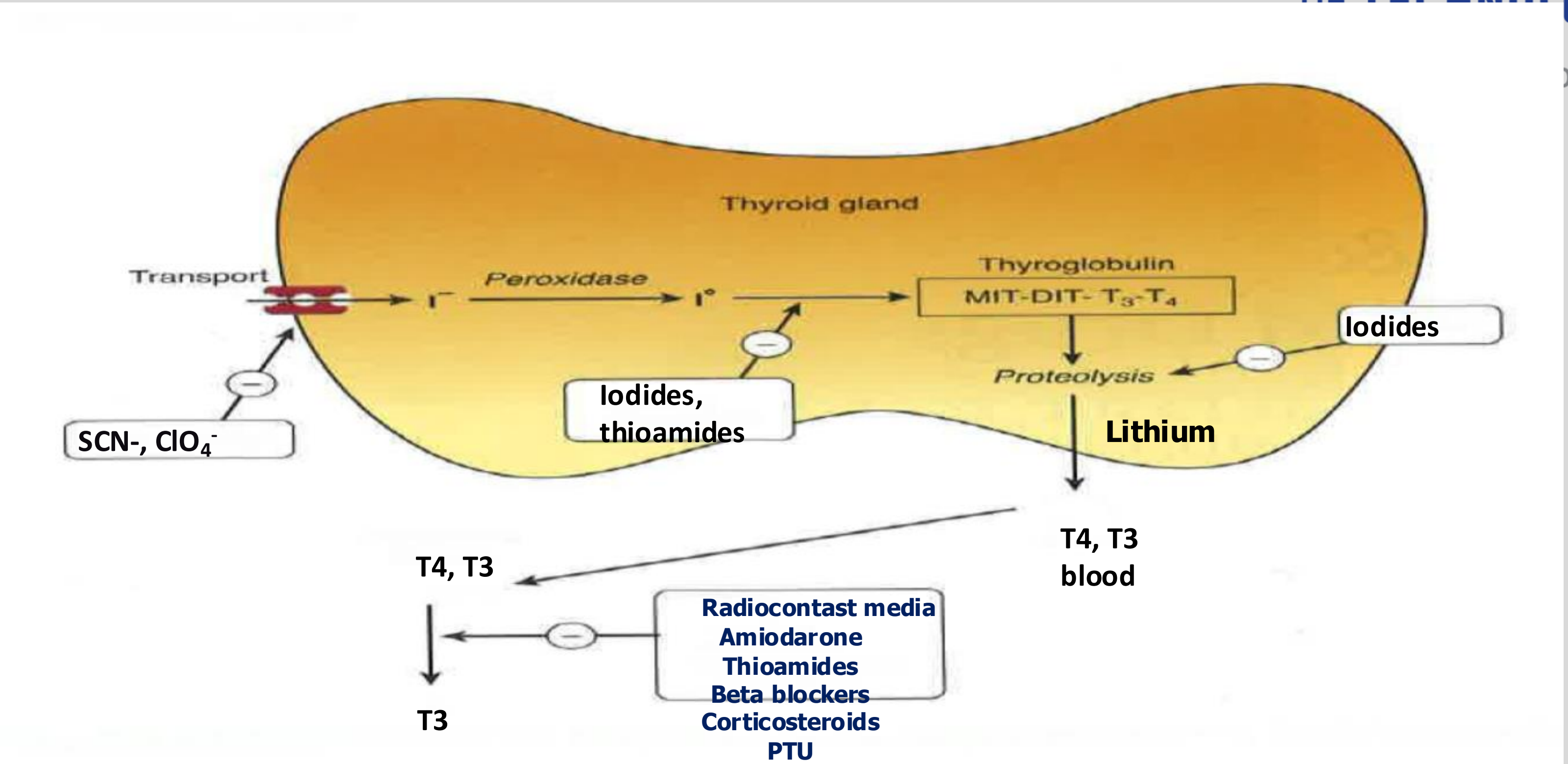
Lithium



Source: Bertram G. Katzung, Anthony J. Trevor: Basic & Clinical Pharmacology, 13th Ed.
www.accesspharmacy.com

Source: Course Syllabus

Iodine Organification



From: Thyroid & Antithyroid Drugs
Basic & Clinical Pharmacology, 14e, 2018

Valproic Acid

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- <u>**Broad-spectrum Mechanisms**</u>

1. **Blocks Na+ channels and T-type calcium channels**
2. **Decrease glutamate at NMDA receptors**
3. **Increases GABA receptor action**
4. **Increases GABA synthesis**
5. **Blocks degradation of GABA**

Source: Course Syllabus

Valproic Acid

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- Use: Tonic-clonic, absence, and partial seizures, migraine prophylaxis, alternative to lithium in bipolar mania.
- <u>Adverse effects:</u> "Valproate Syndrome" includes spina bifida, neural tube defects, autism.
- Weight gain
- Hepatitis
- Inhibitor of CYP 2D6 and 3A4

Source: Course Syllabus

Carbamazepine

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- Blocks sodium channels, blocks NE reuptake
- Use: Tonic-clonic and partial seizures, trigeminal neuralgia, bipolar disease
- Adverse effects: Drowsiness, ataxia, diplopia
- SIADH (water “intox”)= hyponatremia
- Aplastic anemia
- Teratogen (neural tube defects)
- Stevens-Johnson syndrome (HLA B-1502 allele in ethnic population).
- Inducer of CYP 3A4 (also phenytoin, phenobarbital, rifampin, griseofulvin).

Source: Course Syllabus

Lamotrigine

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- **Potentiates GABA, blocks voltage-gated Na+ channels, glutamate blockade.**
- **Use: Partial, tonic-clonic, and absence seizures, manic phase of bipolar syndrome, Lennox-Gastaut syndrome**
- **<u>Adverse effects:</u> sedation, Black Box Warning Stevens-Johnson syndrome= rash**

Source: Course Syllabus

Atypical Antipsychotics (Second generation)

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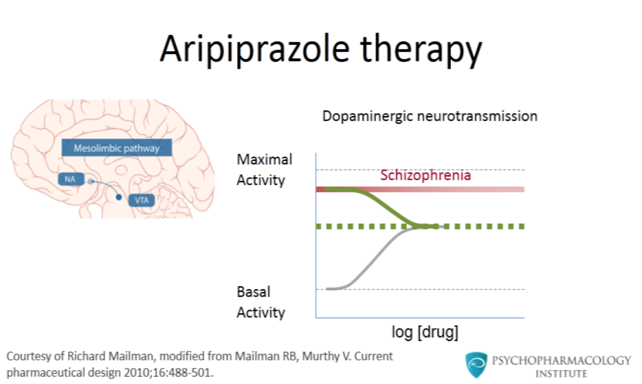
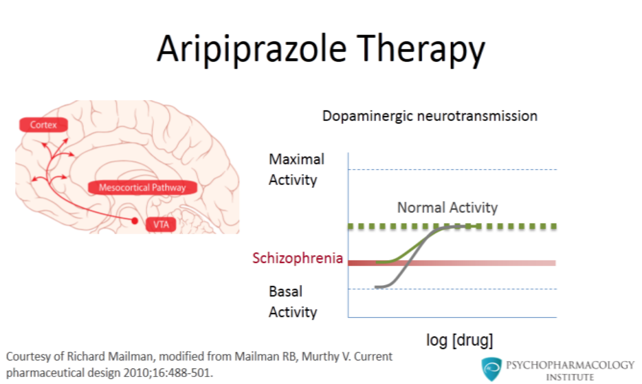
- 5-HT_2 >>>D_2; D_3 and D_4 block
- Increase DA release to relieve negative symptoms (positive too) of psychosis
- Few extrapyramidal effects (EPS) and hyperprolactinemia
- All prolong QT interval. Change in metabolic panel.
- Aripiprazole, olanzapine, quetiapine, risperidone, ziprasidone, and cariprazine, approved for mania

Source: Course Syllabus

Aripiprazole

- Partial D_2 and $5\text{-}HT_{1a}$ agonist, $5\text{-}HT_{2A}$ antagonist.
- Use: bipolar disease, schizophrenia, Tourette's and irritability in autism.
- $5\text{-}HT_{2C}$ agonist (minimal weight gain).
- CYP substrate





Source: Course Syllabus

Atypical Antipsychotics

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- Olanzapine: Less ANS effects, Blocks D_3, D_4, Fewer EPS, *weight gain*, sedation (H_1). Approved in combination with fluoxetine for mania.
    - Receptor blockade: $5\text{-}HT_{2A} > H_1 > D_4 > D_2 > \alpha_1 > D_1$
- Quetiapine:D_2 blocker, which binds for a short time. Minimal muscarinic block, but most sedation (H_1) with hypotension (alpha$_1$).
    - Receptor blockade: $H_1 > \alpha_1 > M_{1,3} > D_2 > 5\text{-}HT_{2A}$

Source: Course Syllabus

Atypical Antipsychotics

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- **Risperidone: Increase prolactin, hyperlipidemia, hyperglycemia**
  - H_1 blockade (sedation, weight gain)
  - $Alpha_1$ block (hypotension)
- **Ziprasidone: blocks 5-HT and NE reuptake.**
  - Skin reaction, eosinophilia
  - H_1 blockade (sedation, weight gain)
  - $Alpha_1$ block (hypotension)
- **Cariprazine: D_2, D_3 partial agonist (5-HT_{1A} agonist)**
  - Akathisia

Source: Course Syllabus

Lumateperone (Caplyta®)

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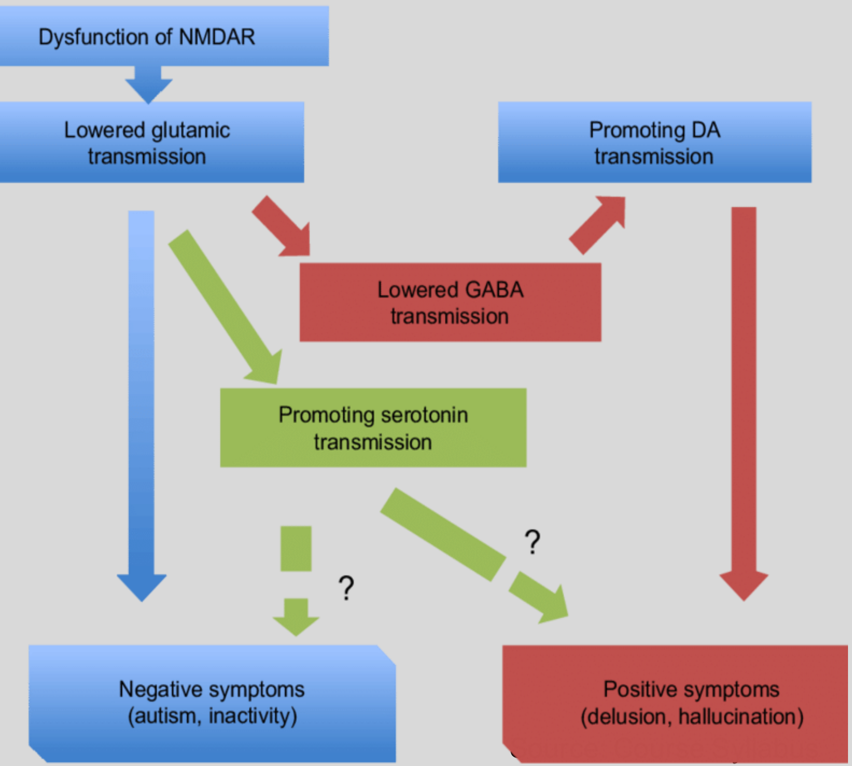
Glutamate hypothesis of schizohrenia



- Use: schizophrenia, MDD, bipolar I and II
- $5\text{-}HT_{2A}$ receptor antagonist (60x greater affinity for this receptor vs D_2 receptors).
- D_2 receptor presynaptic partial agonist and postsynaptic antagonist <u>"Balance"</u>
- D_1 receptor–dependent modulator of glutamate. Augment NMDA receptors

Esketamine (Spravato®)

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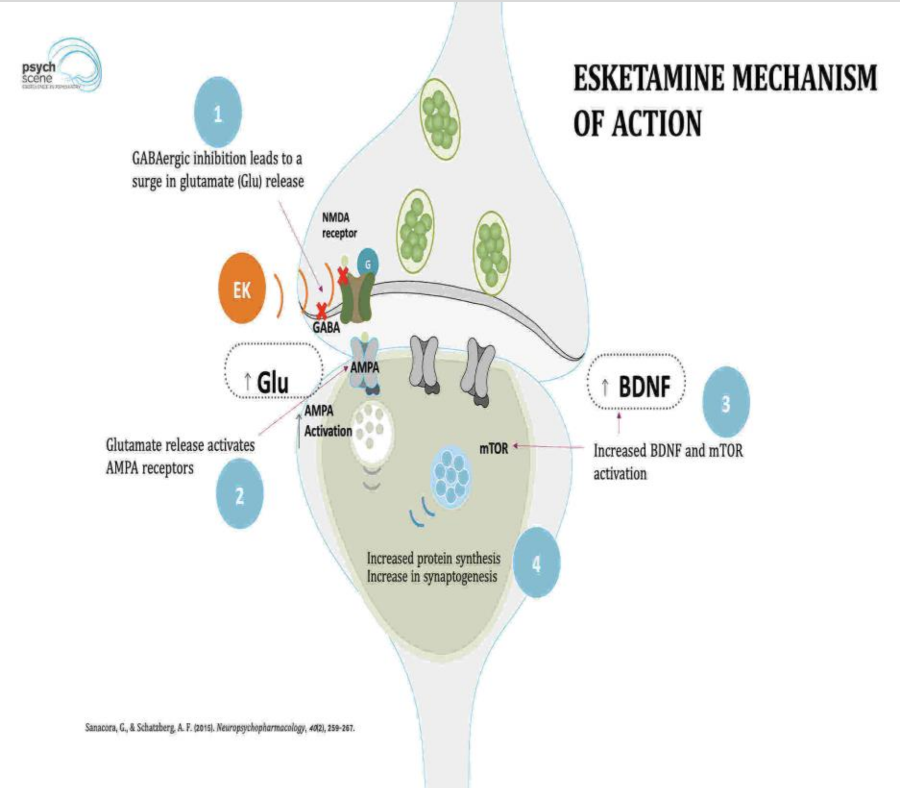

Source: Course Syllabus

- Use: treatment resistant depression
- MOA: NMDA receptor blockade
- <u>Intranasal,</u> administer with oral antidepressant
- Modulates dopamine. Increases BDNF
- Administered in a supervised, clinical setting. Check vitals over 2 hours.
- **AE: Abuse liability, sedation, dissociative effects**

Summary

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- Contrast the various groups of antidepressants. Differentiate between their MOA and adverse effects.
- Review the specific treatments for bipolar disorder.

<u>Core References:</u>

1. Basic & Clinical Pharmacology, 16e, 2024. Bertram G. Katzung, Todd Vanderah, Chapter 29, Antipsychotics and Lithium and Chapter 30, Antidepressants
2. Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e, 2023. Chapter 18: Drug Therapy of Depression and Anxiety Disorders; Chapter 19: Pharmacotherapy of Psychosis and Mania
3. Antidepressants and Mood Stabilizers Summary and Practice Questions (Maria A. Pino, Ph.D).

Source: Course Syllabus

Lecture Feedback Form:

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R